EXPERIMENT 10: SIMULATION OF CRC (CYCLIC REDUNDANCY CHECK) USING C

PROGRAM:

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#include <stdio.h>
#include <string.h>
int main(void) {
    char data[100], divisor[30], temp[140], codeword[140];
    int n, m, i, j, error = 0;
    printf("Enter data bits: ");
    scanf("%99s", data);
    printf("Enter generator bits (e.g. 1011): ");
    scanf("%29s", divisor);
    n = (int)strlen(data);
    m = (int)strlen(divisor);
    if (m < 2 || m > n + 1 || divisor[0] != '1') {
        printf("Invalid generator or generator length.\n");
        return 1;
    }
    for (i = 0; i < n; i++) {
        if (data[i] != '0' && data[i] != '1') {
            printf("Data must contain only 0 and 1.\n");
            return 1;
        }
    }
    for (i = 0; i < m; i++) {
```

```
    if (divisor[i] != '0' && divisor[i] != '1') {
        printf("Generator must contain only 0 and 1.\n");
        return 1;
    }
}
strcpy(temp, data);
for (i = 0; i < m - 1; i++)
    temp[n + i] = '0';
temp[n + m - 1] = '\0';
for (i = 0; i < n; i++) {
    if (temp[i] == '1') {
        for (j = 0; j < m; j++)
            temp[i + j] =
                (temp[i + j] == divisor[j]) ? '0' : '1';
    }
}
strcpy(codeword, data);
strncat(codeword, temp + n, m - 1);
printf("\nOriginal data : %s\n", data);
printf("Generator     : %s\n", divisor);
printf("CRC remainder : %.*s\n", m - 1, temp + n);
printf("Codeword      : %s\n", codeword);
strcpy(temp, codeword);
for (i = 0; i < n; i++) {
```

```
        if (temp[i] == '1') {
            for (j = 0; j < m; j++)
                temp[i + j] =
                    (temp[i + j] == divisor[j]) ? '0' : '1';
        }
    }
    for (i = n; i < n + m - 1; i++)
        if (temp[i] != '0')
            error = 1;
    if (error)
        printf("Receiver: Error detected.\n");
    else
        printf("Receiver: No error detected by CRC.\n");
    return 0;
}
```

OUTPUT :

